SPRING FRAMEWORK

Spring is a framework : (set of library / dependencies / binary / common structure)
Spring Fundamentals :

1)Spring is a comprehensive enterprise application framework for java
2)It provides a container for managing objects through DI
(Dependency Injection) and AOP (Aspect-Oriented Programming)
3) It simplifies application development for the developers with various modules
such as Spring Boot, Spring MVC , Spring Data , Spring Security

Modules Purpose
Spring Core Dependency Injection(Ioc Containers)
Spring MVC Web Applications & Rest APIs(http://localhost:8080/swiggy)
Spring Boot Auto-configured framework for rapid development
Spring AOP Aspect Oriented Programming(Logging , security)
Spring Security Authentication & Authorization
Spring Data simplifies database access (JDBC , JPA , Hibernate)
Spring Cloud Microservices & Distributed Systems

Install prerequisite
-- JDK 17 or higher version
-- Maven (for adding dependencies (just like in core java(we were adding jar files and setting in a class path))
  -- Maven is a build tool (build your maven project after taking few / required configuration)
  -- Gradle
-- IDE (Intellij Idea / Eclipse / VS Code)

-- Postman / Swagger -- To test the APIs or Endpoints

To create a spring Project --
-- create a maven project / gradle -- Which is a building tool
-- After creating a spring project using a maven build tool ,
-- Then we have to add all the required

dependencies in your pom.xml (A main configuration file of your project) from which spring container will take
all the dependencies and set in your class path (Build Path)

Q from where we will add all the dependencies
Ans: Mavenrepository.com is the url and we also call it as central repository

WHAT IS SPRING?

Spring = Java framework

A framework means:

ready-made structure + libraries + tools

that helps developers build applications faster.

SIMPLE REAL LIFE EXAMPLE

Without Spring:

You build a house from raw bricks manually.

With Spring:

You already get doors, windows, pipes, structure ready.

You only focus on business logic.

THIS LINE

Spring is a comprehensive enterprise application framework for Java

Means:

Spring is used for:

- large applications
- company-level software

- web apps
- APIs
- banking systems
- e-commerce

etc.

WHAT IS FRAMEWORK?

Framework = collection of:

- libraries
- dependencies
- predefined structure
- tools

Example:

Spring gives:
- database support
- security
- web support
- API support
- dependency management

already ready.

WHAT IS DEPENDENCY?

Dependency means:

extra code/library needed in project

Example:

If you want MySQL support:

you need MySQL connector jar.

That is dependency.

OLD WAY (CORE JAVA)

Earlier:

download jar manually
→ add to build path

Very irritating.

SPRING WAY

Now Maven downloads automatically.

You only write in:

pom.xml

IOC / DEPENDENCY INJECTION

This is MOST IMPORTANT concept.

WITHOUT SPRING

Suppose:

Student s = new Student();

You are manually creating object.

WITH SPRING

Spring creates object for you.

You only ask:

"Spring give me Student object"

This is:

Dependency Injection (DI)

SIMPLE MEANING

Dependency Injection means:

Spring automatically creates and manages objects

instead of programmer manually creating them.

IOC CONTAINER

IOC means:

Inversion Of Control

Very simple meaning:

control of object creation goes to Spring

instead of programmer.

EXAMPLE

Without Spring:

```
Engine e = new Engine();
Car c = new Car(e);
```

With Spring:

Spring creates Engine
Spring injects into Car automatically

AOP (Aspect Oriented Programming)

Used for:

- logging
- security
- transactions

REAL EXAMPLE

Suppose every function needs:

"User logged in"

Instead of writing again and again:

Spring AOP automatically runs common code.

SPRING MODULES

Spring is BIG.

So it is divided into modules.

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1. SPRING CORE

Main heart of Spring

Provides:

- IOC
- Dependency Injection

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2. SPRING MVC

Used for:

web applications and REST APIs

Example:

http://localhost:8080/swiggy

MVC means:

M Model

V View

M Model

C Controller

FLOW OF MVC

User Request
↓
Controller
↓
Service
↓
Database
↓
Response

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3. SPRING BOOT

Most popular.

Used for:

fast development

It automatically configures things.

Without Boot:

huge configuration.

With Boot:

minimal setup.

EXAMPLE

Without Spring Boot:

100 lines configuration

With Boot:

5 lines

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4. SPRING SECURITY

Used for:

- login
- authentication
- authorization

AUTHENTICATION

Means:

Who are you?

Example:

username/password checking.

AUTHORIZATION

Means:

What are you allowed to access?

Example:

admin panel access.

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5. SPRING DATA

Used for database work.

Supports:

- JDBC
- Hibernate
- JPA

Makes database coding easier.

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6. SPRING CLOUD

Used in:

microservices
distributed systems

Big company systems.

WHY JDK 17?

Spring Boot latest versions require:

Java 17+

WHAT IS MAVEN?

Maven is:

Build Tool

WHAT BUILD TOOL DOES?

It:

- downloads dependencies
- manages libraries
- compiles project
- builds project
- creates jar/war

WITHOUT MAVEN

You manually:

- download jars
- add build path
- manage versions

Very hard.

WITH MAVEN

Only write:

<dependency>

inside:

pom.xml

Maven downloads automatically.

WHAT IS pom.xml?

Main configuration file.

Contains:

- dependencies
- project info
- plugins
- versions

MAVEN REPOSITORY

Website:

[Maven Repository]

Contains millions of dependencies.

FLOW

Suppose you want MySQL dependency.

STEP 1

Search:

mysql connector

on Maven Repository.

STEP 2

Copy dependency code.

STEP 3

Paste into:

pom.xml

STEP 4

Maven downloads jar automatically.

POSTMAN / SWAGGER

Used for API testing.

POSTMAN

Tool to send HTTP requests.

Example:

GET /users
POST /login

SWAGGER

Automatically generates API documentation.

COMPLETE SPRING FLOW

Create Spring Boot project
↓
Add dependencies in pom.xml
↓
Maven downloads libraries
↓
Spring container starts
↓
Spring creates objects automatically
↓
You build APIs/web app quickly

MOST IMPORTANT CONCEPTS

Concept	Simple Meaning
Spring	Java framework
Framework	ready-made structure
Dependency	external library
Maven	dependency manager/build tool
pom.xml	project configuration file
DI	Spring creates objects
IOC	control goes to Spring
Spring Boot	rapid development
MVC	web architecture

Concept	Simple Meaning
AOP	common functionality handling

Tight Coupling

Simple meaning:

Ek class directly dusre specific class par dependent hai.

Example:

private TeamLead teamlead = new TeamLead();

Yaha Delegate class directly bol raha hai:

"Mujhe sirf TeamLead hi chahiye"

Problem Kya Hai?

Kal agar company bole:

TeamLead nahi Manager use karo

toh tumhe:

- code change karna padega
- Delegate class modify karni padegi

Means:

classes strongly connected hain

Isi ko bolte hain:

Tight Coupling

Real Life Example

Suppose TV sirf ek hi company ka cable support karta hai.

Agar cable company badal gayi:

TV todna/modify karna padega

Bad design.

Loose Coupling

Yaha class directly kisi specific class par depend nahi karti.

Instead:

interface par depend karti hai

Example:

private Allocator allocator;

Ab Delegate ko farak nahi padta:

- Manager aaya
- TeamLead aaya
- HR aaya

Bas uske paas:

taskAllocation()

method hona chahiye.

Real Life Example

Mobile charging cable.

Phone bolta hai:

"Mujhe bas Type-C chahiye"

Kaunsi company ka charger hai:

- Samsung

- OnePlus
- Realme

phone ko farak nahi padta.

Yeh hai:

Loose Coupling

Benefit

Agar implementation badlo:

existing code change nahi karna padta

Flexible + reusable + maintainable.

Ab Spring Kaha Aata Hai?

Even in loose coupling code:

new Manager1()

developer khud object bana raha hai.

Means developer still manually decide kar raha:

kaunsa object use hoga

Spring Kya Bolta Hai?

Spring bolta hai:

"Object tum mat banao."
"Main banaunga."
"Main inject karunga."

Without Spring

Developer manually:

new Manager1()

With Spring

Spring automatically:

- object create karta hai
- object manage karta hai
- inject karta hai

using:

- IOC Container
- Dependency Injection

Spring Ka Main Goal

Tight coupling hatao
Loose coupling lao
Object creation Spring ko do

MOST IMPORTANT ONE-LINER

Tight Coupling

Class khud dependency create karti hai

Loose Coupling

Dependency bahar se aati hai

Spring

Dependency Spring automatically provide karta hai